

Proof that $\sqrt{7}$ is Irrational

Problem. Prove that $\sqrt{7}$ is irrational.

Theorem 1. *The number $\sqrt{7}$ is irrational.*

Proof. We argue by contradiction. The proof will avoid assuming a reduced fraction representation, and we will prove explicitly the divisibility fact that is needed.

First, we prove the following elementary lemma.

Lemma. If n is an integer and $7 \mid n^2$, then $7 \mid n$.

Proof of the lemma. By the division algorithm, every integer n can be written as

$$n = 7t + r$$

for some integers t and r with

$$r \in \{0, 1, 2, 3, 4, 5, 6\}.$$

Then

$$n^2 \equiv r^2 \pmod{7}.$$

The possible values of r^2 modulo 7 are

r	0	1	2	3	4	5	6
$r^2 \bmod 7$	0	1	4	2	2	4	1

Thus $r^2 \equiv 0 \pmod{7}$ only when $r = 0$. Therefore, if

$$7 \mid n^2,$$

then $n^2 \equiv 0 \pmod{7}$, so $r = 0$, and hence

$$n = 7t.$$

So $7 \mid n$. This proves the lemma.

Now suppose, for contradiction, that $\sqrt{7}$ is rational. Then there exist integers a and b with $b \neq 0$ such that

$$\sqrt{7} = \frac{a}{b}.$$

Changing the signs of both a and b if necessary, we may assume $b > 0$.

Consider the set

$$S = \left\{ b \in \mathbb{Z}_{>0} : \text{there exists } a \in \mathbb{Z} \text{ such that } \sqrt{7} = \frac{a}{b} \right\}.$$

This set is nonempty because $\sqrt{7}$ is assumed rational. By the well-ordering principle, S has a least element. Choose $b \in S$ minimal, and choose an integer a such that

$$\sqrt{7} = \frac{a}{b}.$$

Squaring both sides gives

$$7 = \frac{a^2}{b^2}.$$

Since $b \neq 0$, multiplying both sides by b^2 gives

$$a^2 = 7b^2.$$

Therefore

$$7 \mid a^2.$$

By the lemma, $7 \mid a$, so there exists an integer c such that

$$a = 7c.$$

Substituting this into $a^2 = 7b^2$, we get

$$(7c)^2 = 7b^2.$$

Hence

$$49c^2 = 7b^2.$$

Dividing both sides by 7 gives

$$7c^2 = b^2.$$

Thus

$$7 \mid b^2.$$

By the lemma again, $7 \mid b$, so there exists an integer d such that

$$b = 7d.$$

Since $b > 0$, we have $d > 0$, and since $7 > 1$,

$$d = \frac{b}{7} < b.$$

Now

$$\sqrt{7} = \frac{a}{b} = \frac{7c}{7d} = \frac{c}{d}.$$

Thus $d \in S$. But $d < b$, contradicting the minimality of b in S .

Therefore our assumption that $\sqrt{7}$ is rational was false. Hence $\sqrt{7}$ is irrational. $\square$